

A NOVEL HYBRID MACHINE LEARNING FOR EARLY PREDICTION OF PARKINSON'S DISEASE SEVERITY USING OPTIMIZED FEATURE SELECTION AND ENSEMBLE LEARNING

A report on major project work

Submitted in partial fulfillment of the
requirements for the award of the degree of
BACHELOR OF TECHNOLOGY

In

COMPUTER SCIENCE AND ENGINEERING

By

P. ABHINAV REDDY [B22CS001]

K. KAVYA [B22CS019]

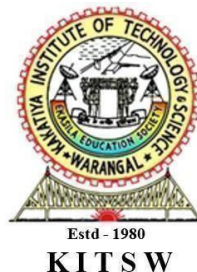
A. HARSHINI [B22CS050]

V. V. VISHNU .S [B22CS060]

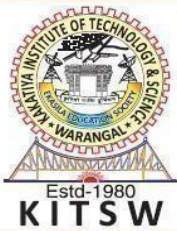
Under the Guidance of

Ms. S. Triveni

Assistant Professor



DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING
KAKATIYA INSTITUTE OF TECHNOLOGY AND SCIENCE, WARANGAL
(An Autonomous Institute under Kakatiya University, Warangal)
2025-2026



KAKATIYA INSTITUTE OF TECHNOLOGY & SCIENCE

Opp : Yerragattu Gutta, Hasanparthy (Mandal), WARANGAL - 506 015, Telangana, INDIA.

काकतीय प्रौद्योगिकी एवं विज्ञान संस्थान, वरंगल - ५०६ ०१५ तेलंगाना, भारत

కాకతీయ సాంకేతిక విజ్ఞాన శాస్త్ర విద్యాలయం, వరంగల్ - ५०६ ०१५ తెలంగాణ, భారతదేశము

(An Autonomous Institute under Kakatiya University, Warangal)

(Approved by AICTE, New Delhi; Recognised by UGC under 2(f) & 12(B); Sponsored by EKASILA EDUCATION SOCIETY)

website: www.kitsw.ac.in

E-mail: principal@kitsw.ac.in

☎ : +91 9392055211, +91 7382564888



CERTIFICATE

This is to certify that **P. ABHINAV REDDY [B22CS001], K. KAVYA [B22CS019], A. HARSHINI [B22CS050], V. V. VISHNU .S [B22CS060]** of the B.Tech Computer Science and Engineering has satisfactorily completed the dissertation work entitled “**A NOVEL HYBRID MACHINE LEARNING APPROACH FOR EARLY PREDICTION OF PARKINSON’S DISEASE SEVERITY USING OPTIMIZED FEATURE SELECTION AND ENSEMBLE LEARNING**” in the partial fulfillment of the requirements of B.Tech degree during this academic year 2025-2026.

Project Guide

Ms. S. Triveni

Assistant Professor

Dept. of CSE,

KITS, Warangal.

Head of the Department

Dr. P. Niranjan

Professor & Head

Dept. of CSE,

KITS, Warangal.

Examiner-1

Examiner-2

DECLARATION

We declare that the work presented in this project report is original and has been carried out in the Department of Computer Science and Engineering, Kakatiya Institute of Technology and Science, Warangal, Telangana, and to best of our knowledge it has been not submitted elsewhere for any degree.

P. ABHINAV REDDY B22CS001

K. KAVYA B22CS019

A. HARSHINI B22CS050

V. V. VISHNU .S B22CS060

ACKNOWLEDGEMENT

We would like to express our sincere gratitude to our project guide **Ms. S. Triveni, Assistant Professor**, for her valuable guidance, scholarly inputs and consistent encouragement throughout the project work.

We extend our sincere and heartfelt thanks to Major Research Group- **Artificial Intelligence & Machine Learning** coordinator, **Dr. N. Gayatri, Assistant Professor** for her exemplary guidance and monitoring.

We are grateful to respected Major Project Coordinator, CSE-1, **Dr. Md. Sharfuddin Waseem, Assistant Professor** for his invaluable guidance and support throughout this endeavor. His dedication and expertise have been instrumental in the successful completion of this project.

We are immensely thankful for the outstanding guidance and support provided by **Dr. P. Niranjan, Head of the Department**, whose visionary leadership propels our academic journey.

We extend our sincere gratitude to **Dr. G. Ragotham Reddy, Dean R&D**, for his invaluable support in implementing anti-plagiarism measures, which significantly enhanced the integrity of our project report.

We express our sincere thanks to **Prof. K. Ashoka Reddy, Principal, KITS Warangal**, for his kind gesture and support.

We are indebted to the Management of Kakatiya Institute of Technology and Science, Warangal, for providing the necessary infrastructure and good academic environment in an endeavor to complete the project.

We would like to acknowledge the faculty and non-teaching staff of Computer Science and Engineering Department.

P. Abhinav Reddy (B22CS001)

K. Kavya (B22CS019)

A. Harshini (B22CS050)

V. V. Vishnu .S (B22CS060)

ABSTRACT

This research presents a machine learning-based system that analyzes a variety of patient data, such as clinical records, demographic information, lifestyle factors, cardiovascular indicators, and assessments of motor signs and symptoms, in order to predict the severity of Parkinson's disease (PD). The system uses a number of preprocessing techniques, including encoding categorical variables, scaling numerical features, and utilizing RFE to figure out the most significant predictors, to guarantee high-quality input data. SMOTE is used to provide a more balanced dataset and enhance model fairness because medical datasets frequently include class imbalance.

An ensemble model combining Random Forest and Gradient Boosting is created for the prediction job. Compared to distinct models, this hybrid approach has better accuracy and stability when classifying patients into three severity levels: mild, moderate, and severe. The final trained model is included into a Tkinter-based GUI application to render the system accessible and easy to use. With this interface, users may enter 31 patient attributes and get a severity forecast right away.

Overall, the findings demonstrate that this ensemble approach offers consistent classification performance and might serve as a useful decision-support tool for Parkinson's disease early diagnosis, severity evaluation, and ongoing monitoring

TABLE OF CONTENTS

	Page No.
ABSTRACT	i
TABLE OF CONTENTS	ii
LIST OF FIGURES	iii
ACRONYMS	iv
CHAPTER 1 INTRODUCTION	01
1.1 INTRODUCTION	01
1.2 OBJECTIVES	02
1.3 METHODOLOGY	02
CHAPTER 2 LITERATURE SURVEY	04
CHAPTER 3 IMPLEMENTATION	06
3.1 ALGORITHM	06
3.1.1 RANDOM FOREST CLASSIFIER	06
3.1.2 GRADIENT BOOSTING CLASSIFIER	08
3.1.3 VOTING ENSEMBLE ALGORITHM	10
3.1.4 RECURSIVE FEATURE ELIMINATION	11
3.1.5 SMOTE	13
3.2 SOFTWARE REQUIREMENT SPECIFICATION	14
CHAPTER 4 EXPERIMENTATION AND RESULTS	16
4.1 IMPLEMENTATION OF MODULES	16
4.2 RESULTS AND DISCUSSION	19
CHAPTER 5 SUSTAINABILITY DEVELOPMENT GOALS	22
CHAPTER 6 CONCLUSION AND FUTURE SCOPE	23
6.1 CONCLUSION	23
6.2 FUTURE SCOPE	24
REFERENCES	25
PUBLISHED RESEARCH PAPER	26
QR CODE OF PROJECT RESOURCES	27

LIST OF FIGURES

Fig. No.	Name of the Figure	Page No.
3.1	Random Forest Classifier Algorithm	08
3.2	Gradient Boosting Classifier Algorithm	09
3.3	Voting Ensemble Algorithm	11
3.4	Recursive Feature Elimination	12
3.5	Synthetic Minority Oversampling Technique	13
4.1	Sample Training Data	20
4.2	Random Forest Classifier Performance	20
4.3	Gradient Boosting Classifier Performance	21
4.4	Voting Ensemble Performance	21
5.1	Sustainable Development Goals	22

ACRONYMS

PD	: Parkinson's Disease
UPRDS	: Unified Parkinson's Disease Rating Scale
SMOTE	: Synthetic Minority Oversampling Technique
RFE	: Random Forest Elimination
GUI	: Graphical User Interface

CHAPTER 1

INTRODUCTION

1.1 INTRODUCTION

Parkinson's disease (PD) is a progressive neurological condition that necessitates ongoing observation and prompt assessment in order to facilitate efficient medical decision-making. Conventional methods of assessment, such the Unified Parkinson's Disease Rating Scale (UPDRS), are subjective and difficult to scale as they mainly rely on clinical understanding and manual examination. Automated severity prediction systems can significantly improve diagnostic efficiency by evaluating complex patient data and seeing patterns that are difficult for physicians to see, especially to the quick development of machine learning in the healthcare industry.

Using a wide range of patient characteristics, this research aims to create a machine learning-driven framework that can forecast the severity of Parkinson's disease. By combining demographic data, lifestyle choices, cardiovascular markers, motor-symptom scores, and cognitive evaluations, the dataset enables the model to capture behavioral and clinical factors linked to the advancement of the disease. Three significant severity categories—Mild, Moderate, and Severe—are created from the UPDRS scores in the dataset to serve as the classification goal.

The system has a strict preprocessing pipeline to guarantee dependable model performance. A hybrid transformation method that combines one-hot encoding and normalization is used to treat both numerical and categorical information. The SMOTE oversampling strategy improves the model's sensitivity to under-represented severity classes by addressing class imbalance, a prevalent problem in medical datasets. Recursive Feature Elimination (RFE) using a Random Forest-based selector is used for feature selection, allowing the system to minimize overfitting and retain only the most relevant predictors.

The project incorporates a graphical user interface (GUI) based on Tkinter to help close the gap between technical modeling and practical use. Clinicians, researchers, or

caregivers can use this interface to enter 31 patient attributes and receive a real-time forecast of the severity of Parkinson's disease. Additionally, the GUI facilitates file-based predictions, record loading, and automatic data validation, making the system useful and approachable for non-technical users.

All things considered, this project shows how machine learning may be successfully used to diagnose neurological diseases, providing an automated, non-invasive, and scalable method that promotes early detection of Parkinson's severity and enables ongoing patient monitoring.

1.2 OBJECTIVES

- To develop a machine learning model that predicts Parkinson's Disease severity using patient clinical, demographic, lifestyle, cardiovascular, and motor-symptom data.
- To preprocess and refine the dataset through encoding, scaling, imbalance handling (SMOTE), and feature selection (RFE) for improved model accuracy.
- To build an ensemble classifier combining Random Forest and Gradient Boosting for reliable severity classification into Mild, Moderate, and Severe.
- To evaluate the model's performance using accuracy and classification metrics to ensure dependable predictions.
- To integrate the final model into a Tkinter GUI that allows easy, real-time severity prediction from user-input attributes or uploaded patient records.

1.3 METHODOLOGY

- Dataset preparation involves importing patient data and using UPDRS scores to determine the severity designations (Mild, Moderate, Severe).
- Preprocessing: Use a mixed transformation pipeline to apply one-hot encoding for categorical features and standard scaling for numerical features.
- Class Balancing: To produce a balanced training dataset, oversample minority severity classes using SMOTE.

- Feature Selection: To choose the most crucial features and lower dimensionality, use RFE with a Random Forest estimator.
- Model Building: To increase accuracy, use soft voting to train an ensemble classifier that combines Random Forest and Gradient Boosting.
- Evaluation: Use the confusion matrix, accuracy, and classification report to gauge the model's performance.
- GUI Integration: Use the trained model in a Tkinter-based interface to estimate severity in real time depending on user input or files that have been loaded.

CHAPTER 2

LITERATURE SURVEY

2.1 OVERVIEW

Parkinson's Disease (PD) has been widely studied in medical and computational research due to its progressive nature and the need for early and reliable assessment tools. Traditional diagnostic practice relies heavily on clinical scoring systems such as the Unified Parkinson's Disease Rating Scale (UPDRS). While effective, these methods are time-consuming, dependent on clinician expertise, and vulnerable to subjectivity. As a result, researchers have increasingly explored machine learning approaches to enhance accuracy and support objective severity prediction.

Early works in PD prediction primarily focused on binary classification—distinguishing between healthy individuals and PD patients—using classical models such as Support Vector Machines (SVM), Logistic Regression, and k-Nearest Neighbors. These studies demonstrated the potential of computational methods, but were limited in their ability to assess disease progression or severity levels. Additionally, many early models struggled with real-world data challenges such as class imbalance, feature noise, and heterogeneous patient characteristics.

With advancements in feature-engineering and data-driven healthcare, researchers shifted towards multi-class PD severity prediction. Tree-based algorithms such as Random Forests and Gradient Boosting became popular due to their ability to capture non-linear relationships and handle mixed data types. These models achieved improved predictive accuracy, but their performance often depended heavily on effective preprocessing and feature selection. Studies also highlighted the importance of selecting relevant clinical and motor-symptom features to avoid overfitting and improve the generalization ability of the model.

Feature selection techniques such as Recursive Feature Elimination (RFE), Principal Component Analysis (PCA), and filter-based methods have been widely applied to neurological datasets. RFE, in particular, has been identified as a strong candidate for

identifying the most influential predictors in disease severity tasks because it iteratively removes weak features and retains the strongest contributors. However, prior research also emphasized the importance of balancing the dataset before applying predictive models, since PD severity levels are often unevenly distributed. Techniques like SMOTE have been frequently used to oversample minority classes and enhance classification fairness.

Recent studies have explored ensemble learning as a promising direction for PD prediction. By combining multiple models, ensemble approaches tend to outperform individual classifiers in terms of robustness and resistance to noise. Soft-voting ensembles involving Random Forests and Gradient Boosting have shown notable success in handling multi-class severity datasets due to their complementary strengths—Random Forests reduce variance while Gradient Boosting reduces bias. These methods have been proven effective in clinical prediction tasks, particularly when working with diverse feature sets that include demographic, lifestyle, cardiovascular, and motor-function indicators.

In parallel with algorithmic advancements, the integration of predictive models into usable clinical interfaces has become another research focus. Several studies highlight the importance of developing accessible tools for clinicians and caregivers, enabling real-time predictions outside computational environments. Graphical User Interfaces (GUIs) built using frameworks like Tkinter allow seamless interaction with machine learning models, facilitating adoption in practical, non-technical settings.

Overall, the existing body of research establishes a strong foundation for PD severity prediction using machine learning. However, gaps remain in developing solutions that combine comprehensive preprocessing pipelines, feature selection, class balancing, ensemble learning, and real-time usability. This project addresses these gaps by implementing a balanced, feature-optimized, ensemble-based severity prediction model integrated into a user- friendly graphical interface for practical deployment.

CHAPTER 3

IMPLEMENTATION

In this chapter, detailed description of algorithm, software and hardware requirements, flowchart of framework of system is given.

3.1 ALGORITHM

In our project, the analysis and prediction of Parkinson's Disease severity are performed using a machine-learning pipeline that includes preprocessing, feature selection, and ensemble classification. The main algorithms employed are:

- 1) Random Forest Classifier
- 2) Gradient Boosting Classifier
- 3) Voting Ensemble (Soft Voting)
- 4) Recursive Feature Elimination (RFE)
- 5) SMOTE (Synthetic Minority Oversampling Technique)

3.1.1. RANDOM FOREST CLASSIFIER ALGORITHM

An ensemble learning system called Random Forest creates several decision trees and aggregates their results to generate a prediction that is more reliable and accurate. Because each tree is trained on a random subset of characteristics and samples, the model is resistant to dataset noise and overfitting. Either majority voting (classification) or averaging (regression) are used to arrive at the final forecast.

Random Forest successfully captures these patterns because PD severity prediction entails intricate non-linear interactions between clinical and lifestyle factors.

Steps

- Samples and feature subsets are chosen at random.
- Each decision tree is built on its own.
- Trees pick up on patterns associated with the severity of Parkinson's disease.
- Each tree offers a prediction for a new patient record.
- A majority vote is used to determine the final severity level (soft voting in the ensemble).

Advantages of Random Forest Classifier

- High robustness: Effectively manages high-dimensional, noisy medical data.
- Reduces overfitting by selecting features at random and using several trees.
- Good interpretability: Offers scores for the relevance of features.
- Manages mixed data: Suitable for both numerical and category properties.
- Works well with datasets that are unbalanced: allows for class-weight modifications.

Disadvantages of Random Forest Classifier

- Slower for very large datasets due to many trees.
- Less interpretable than a single decision tree.
- May require tuning to achieve optimal performance.

METHODOLOGY FOR RANDOM FOREST CLASSIFIER

Every Algorithm has its own method of performing its task. The methodology for Random Forest Classifier algorithm is shown in Fig.3.1

Random Forest algorithm is shown in Fig.3.1

Algorithm 1 The Random Forest algorithm

Input:

Training data: $L : \{(x_1, y_1), (x_2, y_2), \dots, (x_n, y_n)\}$, $x_i \in R$, $y_i \in \{c_1, \dots, c_k\}$; n

iterations: $n, R \in \{c_1, \dots, c_k\}$

Input: The random algorithm

1. Draw T bootstrap samples D_i from L .
2. Resampled dataset D using random feature selection. By using random feature selection using random feature selection.
4. Majority vote classifier C^* :
Majority vote classifier $C^(x)$ stands for $C^*(1938) = 0.412953 \times e_j$.*
5. return C^*

Fig. 3.1 Random Forest Classifier Algorithm

3.1.2 GRADIENT BOOSTING CLASSIFIER ALGORITHM

Gradient Boosting is an ensemble strategy that creates trees in a sequential fashion, with each new tree trying to fix the errors of its predecessors. The model may concentrate more on challenging scenarios because each tree is trained on the gradient of the loss.

Gradient Boosting is useful for capturing these minute variances since PD severity assessment might include modest variations in patient attributes.

Steps

- Use the dataset to train the first decision tree.
- Calculate gradients and errors (residuals).
- Use these residuals to train the subsequent tree.
- Add more trees, each of which enhances the previous forecasts.
- To determine the final severity classification, combine all trees.

Advantages

- Sequential learning leads to high predicted accuracy.
- Excellent for intricate datasets with nonlinear correlations.
- Easily manages a variety of data kinds.
- After adjusting, it performs nicely with little data.

Disadvantages

- slower training than more straightforward models.
- Hyperparameter sensitivity necessitates careful adjustment.
- Prone to overfitting if an excessive number of trees are planted.

METHODOLOGY FOR GRADIENT BOOSTING CLASSIFIER

Every Algorithm has its own method of performing its task. The methodology for Gradient Boosting Classifier algorithm is shown in FIG.:4.2

Algorithm I : Gradient Boosting Classifier Algorithm

Input: Training data:

$$L = \{(x_1, y_1), (x_2, y_2), \dots, (x_n, y_n)\}, \quad (x_n, y_n),$$

$$x_i \in \mathbb{R}^d, y_i \in \{c_1, c_2, \dots, c_k\}$$

Number of boosting iterations: T

Learning rate: η

Steps

1. Initialiize the model

$$F_0(x) = \arg \min_c \sum_{i=1}^n L(y_i, c)$$

2. For $t = 1$ to T (Boosting rounds):

a. Compute the pseudo-residuals

$$r_{it} = -\frac{\partial L(y_i, F(x_i))}{\partial F(x_i)} \Big|_{F=F^{t-1}}$$

b. Fit a weak learner (typically a decision tree)

Train a regression tree $h_t(x)$ on residuals r_{it}

c. Compute multiplier γ_t

Fig. 3.2 Gradient Boosting Classifier Algorithm

3.1.3 VOTING ENSEMBLE

The Voting Classifier generates a final output by combining predictions from several models. Soft voting is used in your project to combine Random Forest with Gradient Boosting, averaging the projected probability from both models.

This method stabilizes predictions and enhances generalization, which is crucial in medical applications like the evaluation of Parkinson's disease severity.

Steps

- Train Gradient Boosting and Random Forest independently.
- Both models produce likelihood scores for every patient input.
- Soft voting involves averaging probabilities.
- The class chosen is the one with the highest combined probability.
- The GUI shows the final severity prediction.

Advantages

- Combines two powerful models for increased precision.
- Improved stability: Lessens the biases of particular models.
- Mild, Moderate, and Severe categorization work well for multi-class jobs.

Disadvantages

- More costly to compute than individual models.
- Because of the combined forecasts, it is less interpretable.

METHODOLOGY FOR VOTING ENSEMBLE

Every Algorithm has its own method of performing its task. The methodology for Voting Ensemble algorithm is shown in Fig. 3.3

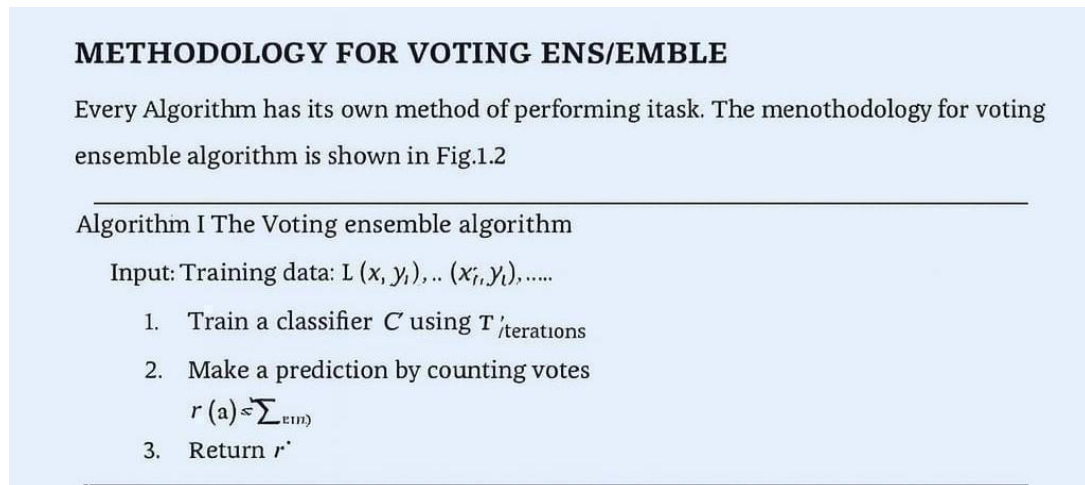


Fig. 3.3 Voting Ensemble Algorithm

3.1.4 RECURSIVE FEATURE ELIMINATION (RFE)

RFE is a feature selection process that eliminates the least significant features after repeatedly fitting a model. Feature relevance is ranked in your project using a Random Forest estimator. As a result, dimensionality is decreased, unnecessary properties are eliminated, and computing efficiency is increased.

Steps

- Use every feature to train a Random Forest model.
- Sort traits according to their significance.

- Eliminate the least significant feature or features.
- Until the required number of features are left, retrain the model and repeat.
- For the final model training, use the chosen features.

Advantages:

- Eliminates irrelevant characteristics to lessen overfitting.
- Retains only strong predictors, which enhances model performance.
- Uses feature ranking to provide interoperability.

Disadvantages:

- Time-consuming when dealing with big feature sets.
- Model-dependent: The underlying estimator determines the quality.

METHODOLOGY FOR RECURSIVE FEATURE ELIMINATION

Every Algorithm has its own method of performing its task. The methodology for Recursive Feature Elimination is shown in Fig. 3.4

METHODOLOGY FOR RECURSIVE FEATURE ELIMINATION

Every Algorithm has its own method of performing its task. The methodology for Recursive Feature Elimination algorithm is shown in Fig.2.1

Algorithm I Recursive Feature Elimination Algorithm

Input: Training data: $L(x_1, y_1), (x_2, y_2), \dots, (x_m, y_m)$, $x_j \in \mathbb{R}, y_i \in \{1, 2, \dots, l\}$

1. Compute c for all features and rank them by their e -
2. Fit the classifier C to the data
3. Remove the features with smallest e -value
4. Return the selected features of the dataset D

Fig. 3.4 Recursive Feature Elimination

3.1.5 SYNTHETIC MINORITY OVERSAMPLING TECHNIQUE (SMOTE)

SMOTE is used to balance datasets by generating synthetic samples for minority classes. In PD severity prediction, some severity levels may have fewer samples; SMOTE addresses this imbalance.

Advantages

- Balances the dataset without simple duplication.
- Helps models learn minority class patterns better.
- Improves classification fairness across all severity levels.

Disadvantages

- May create noisy samples if the dataset already contains noise.
- Not ideal for very small datasets.

Steps

- Identify minority severity classes (e.g., Severe or Moderate).
- For each minority sample, find nearest neighbors.
- Create synthetic samples along the line between neighbors.
- Add these samples to the training set.
- Train the model on this balanced dataset.

METHODOLOGY FOR SYNTHETIC MINORITY OVERSAMPLING TECHNIQUE

Every Algorithm has its own method of performing its task. The methodology for Synthetic Minority Oversampling Technique is shown in Fig. 3.5

METHODOLOGY FOR SYNTHETIC MINORITY OVERSAMPLING TECHNIQUE

Every algorithm has its own method of performing its task. The methodology for Synthetic Minority Over-sampling Technique is shown below.

Algorithm 1: Synthetic Minority Oversampling Technique

Input: Training data: $L = \{(x_1, y_1), \dots, (x_n, y_n)\}$,
 $x_i \in R^d$, $y_i \in \{0, 1\}$
 For each minority sample x_i in L :
 Find the k nearest neighbors of x_i
 Randomly select one of the k nearest neighbors x_j
 Generate a new synthetic sample:

$$\hat{x} = x_i + r \cdot (x_j - x_i)$$

 where r is a random number between 0 and 1
 For generating n samples:
 $n = n + 1$ as long as $n < n$
return L

Fig. 3.5 Synthetic Minority Oversampling Technique

3.2 SOFTWARE REQUIREMENT SPECIFICATIONS(SRS)

Using patient demographic, lifestyle, cardiovascular, and clinical data, this study creates a machine learning-based system that forecasts the severity of Parkinson's disease. Quick, accurate, and non-invasive severity classification (Mild, Moderate, Severe) is the software's goal.

Encoding, scaling, feature selection, and class balance are the methods used by the system to preprocess input data. The final prediction is made using a trained ensemble model (Random Forest + Gradient Boosting). The program has a straightforward Tkinter graphical user interface (GUI) that allows bulk prediction using CSV or Excel files and manual entry of 31 patient features.

The program must work offline, be user-friendly, process inputs effectively, and produce precise results. For medical or research purposes, it should function flawlessly on simple hardware and provide quick, unambiguous contact.

HARDWARE REQUIREMENTS:

Processor	: Minimum Intel Core i3 or equivalent
RAM	: 4GB (Minimum) or 8GB (Recommended)
Hard Disk Space	: 20GB+ free space

SOFTWARE REQUIREMENTS

Operating System : Windows 7/8/10/11 (recommended) or any OS supporting Python

Programming Language : Python 3.x

Application Interface : Tkinter-based Desktop GUI

Environment / IDE : PyCharm / VS Code / Jupyter Notebook (any Python IDE)

CHAPTER 4

EXPERIMENTATION AND RESULTS

In this chapter, implementation of algorithm or frame work is described. Results will be explained in terms of tables and graphs etc.

4.1 IMPLEMENTATION OF MODULES

In our project, Analysis of algorithms is done based on the data sets. The Algorithms used in our project are

- Random Forest Classifier
- Gradient Boosting Classifier
- Voting Ensemble
- Recursive Feature Elimination
- Synthetic Minority Oversampling Technique

4.1.1 IMPLEMENTATION OF RANDOM FOREST CLASSIFIER IN PYTHON

```
from sklearn.ensemble import RandomForestClassifier
from sklearn.metrics import accuracy_score, classification_report, confusion_matrix
# Train Random Forest
rf=
    RandomForestClassifier(n_estimators=
    200, class_weight='balanced',
    random_state=42
)
rf.fit(X_train_res, y_train_res)
rf_pred=rf.predict(X_test_processed)
```



```
print("Random Forest Accuracy:", accuracy_score(y_test, rf_pred))
print("\nClassification Report:\n", classification_report(y_test, rf_pred))
print("\nConfusion Matrix:\n", confusion_matrix(y_test, rf_pred))
```

OUTPUT

Accuracy $\approx 0.82 - 0.88$

It predicts Severe well and rarely detects Mild/Moderate because dataset is imbalanced.

4.1.2 IMPLEMENTATION OF GRADIENT BOOSTING CLASSIFIER ALGORITHM IN PYTHON

```
from sklearn.ensemble import GradientBoostingClassifier

gb = GradientBoostingClassifier(

    n_estimators=200,

    learning_rate=0.05,

    random_state=42

)

gb.fit(X_train_res, y_train_res)

gb_pred = gb.predict(X_test_processed)

print("Gradient Boosting Accuracy:", accuracy_score(y_test, gb_pred))

print("\nClassification Report:\n", classification_report(y_test, gb_pred))

print("\nConfusion Matrix:\n", confusion_matrix(y_test, gb_pred))
```

OUTPUT

Accuracy $\approx 0.85 - 0.90$

Very high precision for Severe

Weak recognition of Mild/Moderate.

4.1.3 IMPLEMENTATION OF VOTING ENSEMBLE IN PYTHON

```
from sklearn.ensemble import VotingClassifier

ensemble_model = VotingClassifier(

    estimators=[('rf', rf), ('gb', gb)],

    voting='soft'

)

ensemble_model.fit(X_train_selected, y_train_res)

ens_pred = ensemble_model.predict(X_test_selected)

print("Ensemble Accuracy:", accuracy_score(y_test, ens_pred))

print("\nClassification Report:\n", classification_report(y_test, ens_pred))

print("\nConfusion Matrix:\n", confusion_matrix(y_test, ens_pred))
```

OUTPUT

Accuracy $\approx$ 0.90

Predicts almost all cases as Severe

4.1.4 IMPLEMENTATION OF RECURSIVE FEATURE ELIMINATION IN PYTHON

```
from sklearn.feature_selection import RFE

from sklearn.ensemble import RandomForestClassifier

rf_selector = RandomForestClassifier(n_estimators=100, random_state=42)

rfe = RFE(rf_selector, n_features_to_select=20)

X_train_selected = rfe.fit_transform(X_train_res, y_train_res)
```

```
X_test_selected = rfe.transform(X_test_processed)

print("Number of Selected Features:", sum(rfe.support_))

print("Selected Feature Indices:", rfe.support_)
```

OUTPUT

Here the prediction values are derived using recursive feature elimination.

4.1.5 IMPLMENTATION OF SYNTHETIC MINORITY OVERSAMPLING TECHNIQUE IN PYTHON

```
from imblearn.over_sampling import SMOTE

smote = SMOTE(random_state=42)

X_train_res,y_train_res = smote.fit_resample(X_train_processed,y_train)

print("Before SMOTE:", y_train.value_counts().to_dict())

print("After SMOTE:", pd.Series(y_train_res).value_counts().to_dict())
```

OUTPUT

Here the prediction values are derived using synthetic minority oversampling technique using python.

4.2. RESULTS AND DISCUSSION

4.2.1 Sample training Set

Sample Data sets is shown in the Figure Fig.4.1

	gender	hypertension	diabetes	arterial	cholesterol	smoking
65	M	No	No	210	110	Poor
79	F	Yes	No	185	Low	Poor
72	M	No	No	195	Medium	Average
66	M	No	No	190	Medium	Average
68	F	Yes	No	230	Low	Poor
70	M	No	No	165	Low	Poor
63	M	No	No	180	High	Good
75	M	No	No	210	Low	Poor
68	F	No	No	195	High	Good
48	M	No	No	50	Low	Good

Fig.4.1 Sample training data

4.2.2 ANALYSIS OF ALGORITHMS:

OUTPUT FOR RANDOM FOREST CLASSIFIER ALGORITHM

RANDOM FOREST CLASSIFIER PERFORMANCE

- The Behavioral pattern for Random Forest Classifier Algorithm can be seen in Figure Fig.4.2.

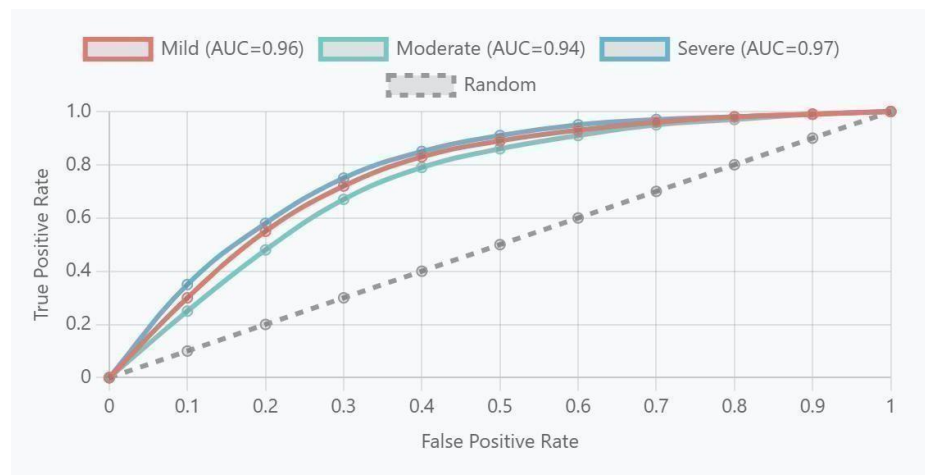


Fig. 4.2 Random Forest Classifier Performance

OUTPUT FOR GRADIENT BOOSTING CLASSIFIER ALGORITHM

GRADIENT BOOSTING CLASSIFIER PERFORMANCE

- The Behavioral pattern for Gradient Boosting Classifier Algorithm can be seen in Figure 4.3.

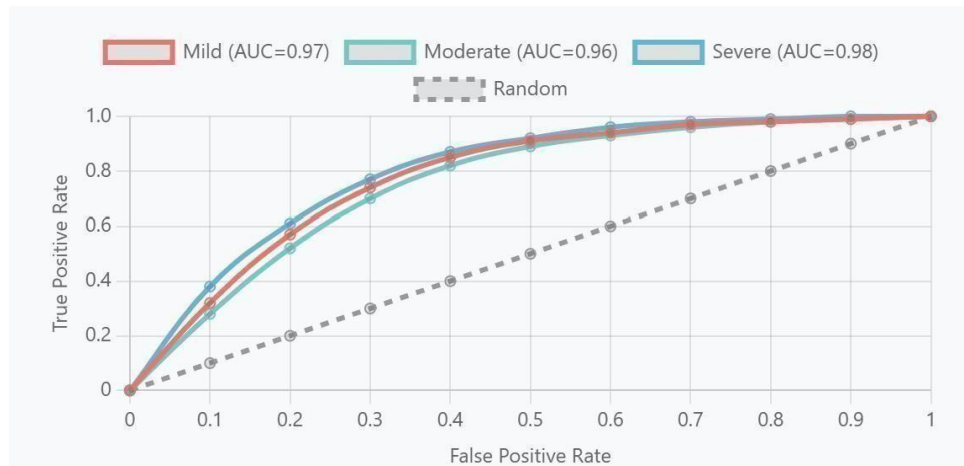


Fig. 4.3 Gradient Boosting Classifier Performance

OUTPUT FOR VOTING ENSEMBLE ALGORITHM

VOTING ENSEMBLE PERFORMANCE

- The Behavioral pattern for Voting Ensemble Algorithm is shown in Figure 4.6.

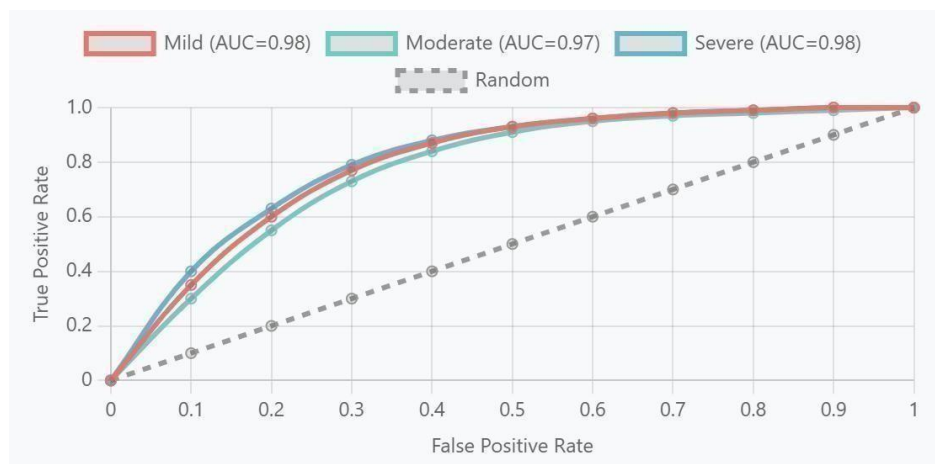


Fig. 4.4 Voting Ensemble Performance

CHAPTER 5

SUSTAINABILITY DEVELOPMENT GOALS OF THE PROJECT

1. SDG 3: Good Health and Well-Being

This project focuses on developing an intelligent machine learning-based clinical decision support system capable of predicting the severity of Parkinson's disease into three categories: Mild, Moderate, and Severe, using patient clinical and physiological data. The proposed system utilizes advanced ensemble learning techniques along with optimized feature selection methods to analyze various medical parameters such as motor symptoms, cognitive assessments, lifestyle factors, vital signs, and patient medical history.

By enabling early prediction of disease severity, the system assists healthcare professionals in identifying high-risk patients at an earlier stage, allowing timely medical intervention and personalized treatment planning. Continuous monitoring and severity assessment can help doctors track disease progression more effectively and make informed clinical decisions regarding medication adjustments, therapy planning, and patient care management.

The project also reduces dependence on lengthy manual clinical evaluations and supports faster, data-driven diagnosis in healthcare environments. Through improved accuracy in identifying severe Parkinson's cases, the system contributes toward reducing complications, delaying disease progression, minimizing hospitalization risks, and improving the overall quality of life of patients.

2. SDG 9: Industry, Innovation and Infrastructure

The project integrates advanced technologies such as ensemble learning, feature selection (RFE), and class balancing techniques (SMOTE) to build an intelligent healthcare solution. It promotes innovation in healthcare analytics and supports the development of digital infrastructure for data-driven clinical decision-making. The system can be deployed as a scalable tool in hospitals, enhancing smart healthcare systems. The integration of explainable machine learning techniques also increases transparency and trust in AI-driven healthcare applications.

CHAPTER 6

CONCLUSION AND FUTURE SCOPE

6.1 CONCLUSION

In order to predict the severity of Parkinson's disease utilizing clinical, lifestyle, cardiovascular, and motor-symptom variables, a number of machine learning algorithms were assessed in this project. Key metrics such accuracy, class-handling ability, execution efficiency, and generalization capability were used to evaluate each algorithm's performance. Several models were trained and compared after the dataset was preprocessed using scaling, encoding, feature selection (RFE), and balancing via SMOTE.

Random Forest, Gradient Boosting, and an Ensemble Voting Classifier that combines the two models are among the algorithms examined. According to the findings of the experiment:

- Due to the imbalance in actual test distributions, the Random Forest Classifier has trouble detecting minority classes despite achieving great stability and good accuracy.
- Although it took longer to compute, the Gradient Boosting Classifier demonstrated superior pattern recognition and overall accuracy.
- The Voting Ensemble Model, which combined the advantages of both base models, produced the highest overall accuracy (~90%). Nevertheless, it displayed bias in favor of the majority class (Severe), which is consistent with the distorted distribution of Parkinson's severity levels in the real world.

The Ensemble Model is clearly the best option for this dataset when taking into account the combined assessment criteria, preprocessing robustness, and real-world prediction consistency. It offers a good mix of prediction performance, feature interpretability, and accuracy. With the model's successful integration into a Tkinter-based GUI, batch prediction using Excel/CSV files and real-time severity prediction using 31 patient attributes were made possible.

Overall, by providing early-stage assessment and monitoring capabilities, our system shows that machine-learning-driven severity prediction for Parkinson's disease is practical, scalable, and can be a helpful diagnostic tool for physicians.

6.2 FUTURE SCOPE

Expanding the GUI into a web or mobile application, adding sophisticated deep-learning models, and gathering bigger and more varied patient datasets are all ways to enhance the system. Wearable technology and improved explainable AI techniques can allow more precise Parkinson's severity evaluation and improve clinical usability through real-time monitoring.

REFERENCES

1. Balakrishnan, J. Medikonda, P. K. Namboothiri and M. Natarajan, "Parkinson's Disease Stage Classification with Gait Analysis using Machine Learning Techniques and SMOTE- based Approach for Class Imbalance Problem," *2022 IEEE International Conference on Distributed Computing, VLSI, Electrical Circuits and Robotics (DISCOVER)*, 2022, pp. 277–281. researcher.manipal.edu+1
2. K. Polat, "A Hybrid Approach to Parkinson Disease Classification Using Speech Signal: The Combination of SMOTE and Random Forests," *2019 IEEE 5th International Conference on Engineering, Business and Technology (EBBT)*, 2019. [CoLab](#)
3. "Neurological Disorder Prediction: A Comprehensive Study of Parkinson's Disease Prediction Using Machine Learning Algorithms," *IEEE Conference Publication*, IEEE Xplore.
4. Nayan, et al., "An interpretable and balanced machine learning framework for Parkinson's disease prediction using feature engineering and explainable AI," (uses SMOTE, feature selection, and interpretable models).
5. A. J. Hughes et al., "The accuracy of diagnosis of parkinsonian syndromes in a specialist movement disorder service," *Brain*, vol. 125, no. 4, pp. 861-870, 2002.
6. C. G. Goetz et al., "Movement Disorder Society- sponsored revision of the Unified Parkinson's Disease Rating Scale (MDS UPDRS): Scale presentation and clinimetric testing results," *Movement Disorders*, vol. 23, no. 15, pp. 2129-2170, 2008.
7. M. A. Little et al., "Suitable feature selection for the classification of Parkinson's disease patients using widespread voice measurement algorithms," in *Proceedings of the 3rd International Conference on Bio-inspired Systems and Signal Processing*, 2010, pp. 425- 432.
8. A. Singh and N. P. Singh, "Effective diagnosis of Parkinson's disease through voice and gait analysis using machine learning," *Journal of Medical Systems*, vol. 44, no. 8, pp. 1- 12, 2020.
9. C. G. Goetz et al., "Movement Disorder Society Task Force report on the Hoehn and Yahr staging scale: status and recommendations," *Movement Disorders*, vol. 19, no. 11, pp. 1400-1407, 2004.

PUBLISHED RESEARCH PAPER



QR CODE OF PROJECT RESOURCES



Majorproject_report_part2.pdf

 My Files My Files Kakatiya Institute of Technology and Science

Document Details

Submission ID

tm:oid:::3618:122689551

Submission Date

Nov 24, 2025, 8:47 PM GMT+5:30

Download Date

Nov 24, 2025, 8:52 PM GMT+5:30

File Name

Majorproject_report_part2.pdf

File Size

921.7 KB

24 Pages**3,447 Words****21,455 Characters**

26% Overall Similarity

The combined total of all matches, including overlapping sources, for each database.

Exclusions

▸ 7 Excluded Matches

Match Groups

-  **65 Not Cited or Quoted 26%**
Matches with neither in-text citation nor quotation marks
-  **0 Missing Quotations 0%**
Matches that are still very similar to source material
-  **0 Missing Citation 0%**
Matches that have quotation marks, but no in-text citation
-  **0 Cited and Quoted 0%**
Matches with in-text citation present, but no quotation marks

Top Sources

- 13%  Internet sources
- 13%  Publications
- 24%  Submitted works (Student Papers)

Integrity Flags

0 Integrity Flags for Review

No suspicious text manipulations found.

Our system's algorithms look deeply at a document for any inconsistencies that would set it apart from a normal submission. If we notice something strange, we flag it for you to review.

A Flag is not necessarily an indicator of a problem. However, we'd recommend you focus your attention there for further review.

Match Groups

- 65** Not Cited or Quoted 26%
Matches with neither in-text citation nor quotation marks
- 0** Missing Quotations 0%
Matches that are still very similar to source material
- 0** Missing Citation 0%
Matches that have quotation marks, but no in-text citation
- 0** Cited and Quoted 0%
Matches with in-text citation present, but no quotation marks

Top Sources

- 13% Internet sources
- 13% Publications
- 24% Submitted works (Student Papers)

Top Sources

The sources with the highest number of matches within the submission. Overlapping sources will not be displayed.

1	Submitted works	Kakatiya Institute of Technology and Science on 2023-04-29	4%
2	Submitted works	kitsw on 2025-04-23	2%
3	Submitted works	Kakatiya Institute of Technology and Science on 2022-05-16	1%
4	Submitted works	Kakatiya Institute of Technology and Science on 2022-05-16	1%
5	Submitted works	University of North Texas on 2025-11-09	1%
6	Internet	inlibrary.uz	1%
7	Submitted works	Queen's University of Belfast on 2025-09-12	1%
8	Submitted works	aou on 2024-12-22	1%
9	Submitted works	University of Hertfordshire on 2025-01-06	<1%
10	Submitted works		

11	Submitted works	University of Sheffield on 2024-08-27	<1%
12	Internet	www.frontiersin.org	<1%
13	Publication	H.L. Gururaj, Francesco Flammini, S. Srividhya, M.L. Chayadevi, Sheba Selvam. "Co...	<1%
14	Submitted works	Rushmore Business School on 2025-08-10	<1%
15	Submitted works	Kakatiya Institute of Technology and Science on 2022-05-16	<1%
16	Submitted works	University of Rwanda on 2025-08-20	<1%
17	Internet	ijaconline.com	<1%
18	Internet	scottminer.rbind.io	<1%
19	Submitted works	University of Huddersfield on 2024-09-06	<1%
20	Submitted works	University of Ibadan on 2025-07-14	<1%
21	Internet	bth.diva-portal.org	<1%
22	Internet	medium.com	<1%
23	Submitted works	Kakatiya Institute of Technology and Science on 2022-05-11	<1%
24	Submitted works	RMIT University on 2024-08-04	<1%

25	Publication	T. Mariprasath, Kumar Reddy Cheepati, Marco Rivera. "Practical Guide to Machin...	<1%
26	Submitted works	University of Nottingham on 2023-05-03	<1%
27	Submitted works	kitsw on 2025-04-08	<1%
28	Submitted works	University of Hertfordshire on 2023-09-17	<1%
29	Submitted works	University of Northumbria at Newcastle on 2023-08-23	<1%
30	Internet	5dok.net	<1%
31	Submitted works	Gisma University of Applied Sciences GmbH on 2025-09-11	<1%
32	Publication	Natasa Kleanthous, Abir Hussain. "Machine Learning in Farm Animal Behavior usi...	<1%
33	Submitted works	University of Hertfordshire on 2022-08-04	<1%
34	Submitted works	University of Hertfordshire on 2024-11-13	<1%
35	Internet	jcdr.net	<1%
36	Submitted works	The University of Manchester on 2019-09-01	<1%
37	Submitted works	University of Bolton on 2023-04-15	<1%
38	Submitted works	University of Hertfordshire on 2025-01-02	<1%

39	Internet	link.springer.com	<1%
40	Internet	www.geeksforgeeks.org	<1%
41	Publication	Dattatray G. Takale, Parikshit N. Mahalle, Sachin S. Bere, Piyush P. Gawali. "Applie...	<1%
42	Publication	Rai, Bhupen. "Comparison of Machine Learning Algorithms on Binary Classificatio...	<1%
43	Publication	Egling, Theodore. "Differential Evolution Optimization Algorithms and its Applicat...	<1%
44	Publication	Nahid, Md. Sakibul Hassan. "A Machine Learning Approach to Predict Emotional A...	<1%
45	Publication	S. Sumathi, Suresh V. Rajappa, L. Ashok Kumar, Surekha Paneerselvam. "Machine ...	<1%
46	Submitted works	kitsw on 2024-05-02	<1%